Electric Vehicle Market Performance & Insights

Data Quality Audit

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

The audit covers completeness, validity, consistency, uniqueness, traceability and reconciliation. A cleaned output exists according to the project information, but this alone does not prove every field is correct.

Detailed Project-Specific Content

- Completeness: calculate null/blank rates for every field and identify chart-critical columns.
- Validity: test types and permitted domains for year, battery capacity, range, price and rating.
- Consistency: inspect units, category spellings, manufacturer aliases and year conventions.
- Uniqueness: define record grain and assess duplicate candidates using an appropriate key.
- Plausibility: flag suspicious values for review without assuming all statistical outliers are errors.
- Integrity: reconcile CSV row count with the JSON report and confirm expected columns remain available.
- Traceability: map each cleaning operation to source code and report evidence.
- Visualization consistency: compare Tableau aggregates with independent Pandas/SQL calculations on identical filters.
- Audit table fields: check, rule, observed result, status, evidence file and corrective action.
- Use Pass, Pass with caveat, Fail or Not tested. Critical failures include unexplained count differences, incorrect units, invalid conversions or misleading labels.
- Actual status and statistics: [TO BE POPULATED from cleaning_report.json, code execution and independent checks].

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.